

A Mori–Zwanzig memory kernel for subglacial drainage: the surge-lag memory is hydraulic, not ice-thermal

Abdelrahman Saifelden¹

¹Alexandria Higher Institute of Engineering and Technology (AIET), Alexandria, Egypt

Abstract. Post-drainage ice-stream speed-ups lag their subglacial trigger by months to a couple of years, and the lag carries *memory*: the response rises to a peak rather than decaying monotonically. We make one claim and defend it three ways. **The lumped, linearised cavity↔channel storage system’s surge response is an exact Mori–Zwanzig memory kernel:** projecting out the channel variable yields a generalized Langevin equation whose kernel equals the eliminated subsystem’s own Green’s function (verified to 7×10^{-18} , projection exact to 5×10^{-16} , reduced model reproducing the full trajectory to 9×10^{-8}), with the Markovian (adiabatic-elimination) local closure recovered exactly as the channel time vanishes. This certifies *why* surge response carries memory and *how* it degenerates to a local closure — a structural result, not a field-validated lag value. We motivate it by **falsifying the only zeroth-order alternative:** a literal ice-thermal lag $\tau_{ice} = H^2/\kappa_{ice}$ is rejected by ~ 5 orders of magnitude on the 131 Siegfried & Fricker (2018) active lakes (median $\tau_{ice} \approx 1.5 \times 10^5 yr$ vs observed 0.02–2 yr; robust on the independent Arthern et al. (2015) thickness inversion, $r = 0.970$), and on first principles (the thermal skin depth at surge periods is metres, and an impulse gives a peakless $t^{-1/2}$ response while observations peak). We then expose the kernel to the one open co-temporal dataset — a matched lake-drainage→velocity test (CryoSat-2 drainage dates + ITS_LIVE speeds) — which returns an **honest null** (peak $+0.56\sigma$, 0/5 significant), limited by temporal aliasing and an amplitude bound, not record precision, with two concrete unblock routes. Finally we include, as a continental *screen* rather than a forecast, the Regime Transition Number RTN , show it is by construction the flotation/effective-pressure ratio (so a constant- ϕ RTN adds no skill over thickness-above-flotation), and identify its only genuine degree of freedom — a spatially-varying basal-connectivity fraction $\phi(x, y)$ — with a plant-and-recover proof and a real-Bedmap2 hydraulic-routing demonstration. The contributions are the exact MZ certification (with two extensions that retire the “it is just a 2×2 linear-algebra identity” objection: a *nonlinear* extension — the linear kernel is the small-amplitude limit of the Röthlisberger system, which predicts that bigger floods lag longer; and a *distributed* extension — the kernel is an exact, spatially non-local operator projection of the linearised flowline PDE whose no-transport, single-node limit is exactly the lumped 2×2 , and which predicts a finite along-flow memory footprint), the honest-falsifier discipline, and the diagnosis of the matched-lag null — not a verified surge-lag value.

1 Introduction

Subglacial hydrology controls ice-stream sliding through the effective pressure $N = p_i - p_w$, the difference between ice overburden and basal water pressure (Cuffey & Paterson 2010; Schoof 2010; Röthlisberger 1972). Drainage events — subglacial-

lake floods — perturb N and trigger transient speed-ups (Stearns et al. 2008; Siegfried et al. 2016), but the speed-up *lags* the drainage and the lagged response is non-monotone (it peaks), implying the cavity $\leftrightarrow$ channel system carries memory (Werder et al. 2013; Hewitt 2013). The modelling question is what sets that memory.

30 The mainstream tools are established and we use, not claim, them: the Mori–Zwanzig/Nakajima–Zwanzig projection and the generalized Langevin equation (Mori 1965; Nakajima 1958; Zwanzig 1960, 1973), the Röthlisberger channel and GlaDS-type cavity $\leftrightarrow$ channel drainage (Röthlisberger 1972; Werder et al. 2013; Hewitt 2013), and the flotation/effective-pressure criterion (Schoof 2007; Leguy et al. 2014). What is new here is not a mechanism but an **exact structural certification** and the **discipline applied to its alternatives**.

35 **What is new, stated against the closest prior work.** Werder/Hewitt model the cavity $\leftrightarrow$ channel system forward; we instead *reduce* its linearisation by exact projection and show the surge-lag memory kernel **is** the eliminated channel subsystem’s Green’s function, with the local (Markovian) closure recovered as a limit — a closed-form statement of *why* the lag has memory and *when* a local sliding closure is admissible. No established theory predicts the literal H^2/κ thermal lag; we state it as an explicit straw hypothesis precisely so the active-lake record can reject it, and report that rejection as a *motivated null*
40 rather than a discovery.

Contributions.

1. A **motivated falsification** of the literal H^2/κ ice-thermal sliding-lag kernel on 131 real lakes, relocating the memory into hydrology (§3).
2. The **exact Mori–Zwanzig structure** of the cavity $\leftrightarrow$ channel hydraulic memory kernel — a structural certification (§4),
45 with a **nonlinear extension** showing the linear kernel is the small-amplitude limit of the physical Röthlisberger system and predicting an amplitude-dependent (bigger-floods-lag-longer) surge lag (§4.4), and a **distributed (spatially-resolved) extension** showing the kernel is an exact operator projection of the linearised flowline PDE — the lumped 2×2 is its no-transport, single-node limit, and channel transport gives the surge-lag memory a finite along-flow footprint (§4.5).
- 50 3. An **honest matched-lag null** with a precise account of *why* it is null (aliasing + amplitude bound, not noise) and two routes to a verified lag (§5).
4. A nondimensional **intrusion screen** RTN , shown to equal the flotation ratio, with its only genuine degree of freedom (a data-derived $\phi(x, y)$) isolated and tested (§6) — presented as a screen, not a verified forecast.
5. **Calibrated estimators** via synthetic plant-and-recover (§7).

Dataset	What	Access
BAS Bedmap2	thickness, bed, surface, grounded mask (1 km)	open, no auth
USAP-DC 601470 (Stubblefield et al. 2021)	independent thickness (Arthern et al. 2015) + 131 lake stats	open (USAP-DC)
BedMachine v4 (NSIDC-0756)	independent continental thickness (RTN screen)	NSIDC (Earthdata)
Siegfried & Fricker (2018)	131 active subglacial-lake outlines	open (GitHub mirror)
ITS_LIVE	surface-speed datacubes (image-pair v)	open (anonymous S3)
CryoSat-2 in-lake elevation	drainage <i>dates</i>	open (Siegfried 2021-GRL mirror)
USAP-DC 601439	ICESat lake volume-change (drainage volumes)	open

Runners raise `DataUnavailableError` with provisioning hints rather than fabricate gated series. Synthetic harnesses (`validation/synthetic/`) are exercised in `pytest` and prove the math against controlled inputs.

3 The thermal sliding-lag kernel, falsified

60 **3.1 The hypothesis (a motivated null).** No established sliding theory predicts that surge lag equals the *full-thickness* ice thermal-diffusion time; we state $\tau_{ice} = H^2/\kappa_{ice}$ ($\kappa_{ice} = 1.09 \times 10^{-6} m^2 s^{-1}$) as an explicit, falsifiable straw hypothesis precisely so the data can reject it.

3.2 The test. On real Bedmap2 thickness at the 131 Siegfried & Fricker (2018) catalogued active lakes (130 with valid thickness; median $H = 2282m$, range 637–3905 m):

- 65
- $\tau_{ice} = H^2/\kappa_{ice}$: **median $\approx 1.5 \times 10^5$ yr** ($p5 \approx 2 \times 10^4$; $p95 \approx 3 \times 10^5$ yr).
 - Observed post-drainage surge lags: **0.02–2 yr** (Stearns et al. 2008; Siegfried et al. 2016).
 - $\Rightarrow$ the literal kernel is **$\sim 8 \times 10^4 \times$ too slow** at the median lake; the $\log_{10}\tau_{ice}$ histogram (10^4 – 10^6 yr) is fully disjoint from the observed band.

3.3 Robust to the thickness dataset. Recomputing on an independent ice-thickness dataset (USAP-DC 601470, Stubblefield et al. 2021; ice thickness after Arthern et al. 2015): median thickness 2356 m, τ_{ice} median $\approx 1.55 \times 10^5$ yr; the 601470-vs-Bedmap2 thickness at the same centroids agrees at $r = 0.970$, so the falsification is not an artefact of one sampling.

3.4 Why the falsification is diagnostic. Two facts exclude ice thermal diffusion as the lag-setting mechanism: (i) the thermal skin depth at surge periods is $\delta_{skin} = \sqrt{\kappa_{ice}P/\pi} \approx 0.5$ – $5m$ ($P = 0.02$ – $2yr$) — the perturbation never penetrates beyond metres of ice, so the full-thickness H is the wrong scale; (ii) a drainage is an *impulse*, whose semi-infinite diffusive response is a monotone $t^{-1/2}$ decay with **no peak**, yet observed speed-ups **rise to a peak**. The lag is therefore **hydromechanical**,

75

not thermal — the memory relocates into subglacial hydrology (cavity/channel pressurisation, effective-pressure control of sliding), with the ice-thermal $t^{-1/2}$ term subdominant.

4 The replacement kernel is an exact Mori–Zwanzig memory kernel

We model the **hydraulic subsystem** as cavity↔channel storage–transport (with ϕ the hydraulic potential, *not* the Leray pressure). Its lumped linearisation $\dot{x} = Mx + F$, $x = (s, q)$ (s the resolved cavity store, q the eliminated channel variable, $M = [[-1/\tau_1, -a], [b, -1/\tau_2]]$), reduces by exact projection (linear Nakajima–Zwanzig / Mori, no Markov approximation) to

$$\dot{s}(t) = M_{ss}s(t) + \int_0^t K(t-\tau)s(\tau)d\tau + R(t), \quad K(\tau) = M_{sq}M_{qs}e^{M_{qq}\tau},$$

a non-local (memory) equation whose kernel $K(\tau) = -ab \cdot e^{-\tau/\tau_2}$ is exactly the channel subsystem’s own Green’s function weighted by the up/down couplings. The harness certifies five properties (validation/synthetic/hydraulic_mz_ projection_synthetic.py, 6/6 tests):

1. **Projection exact to machine precision.** The GLE Laplace transfer function equals the full 2×2 resolvent’s (1,1) component for 400 random stable overdamped systems — max abs err 5.0×10^{-16} .
2. **The kernel IS the eliminated subsystem’s Green’s function** — reproduced to 6.9×10^{-18} , decaying at exactly $1/\tau_2$, sign-definite.
3. **The reduced GLE reproduces the resolved trajectory** to rel-err 9.1×10^{-8} over the transient.
4. **Memory is necessary for the lag/peak.** The full response peaks at the interior time set by the coupled eigenvalues ($t^* = 0.911$, analytic 0.911); the Markovian adiabatic-elimination closure collapses to a monotone response with argmax at $t = 0$.
5. **Markovian limit = adiabatic elimination (DC gain).** $\int_0^\infty K d\tau = M_{sq}M_{qs}/(-M_{qq})$ (err 7.5×10^{-9}); shrinking the channel time $\tau_2 \rightarrow 0$ at fixed DC gain collapses the kernel toward $(\int K) \cdot \delta(\tau)$, recovering the local closure — the FDT-Markov limit, here *derived from the projection* rather than assumed.

This certifies the *structure* of the replacement mechanism (that the surge-lag memory is a genuine MZ kernel obtained by projecting out the channel, and how it degenerates to a local closure). (Werder et al. 2013; Hewitt 2013; Schoof 2010; Röthlisberger 1972)

Honest scope. This validates the **projection structure** of the *linearised lumped* model. It does **not** validate the physical identity of ϕ , that this subsystem dominates a real surge, or any field lag value — those remain gated on real co-temporal velocity data (§5).

4.3 Physical magnitudes. The reduced kernel is dimensionless; the bridge to metres uses only known ice constants. With $\rho_i L = 3.0 \times 10^8 \text{ J m}^{-3}$, Glen A , effective pressure N , and literature subglacial inputs (Q , $\partial\phi/\partial s$), the steady R-channel $S^* =$

105 V_o/k_{creep} , $R^* = \sqrt{2S^*/\pi}$, $\tau = 1/k_{creep}$ is **metre-scale** (central $R^* \approx 2.4m$, $\tau \approx 0.18yr$; 73 % of the literature box gives 0.1–50 m) — the R-channel regime, no free fudge factor; the wide upper tail is the low- N near-flotation limit ($k_{creep} \propto N^3 \rightarrow 0$). The roughness leg uses $z_0 = c_z \cdot a$ in the log-law drag $C_d = [\kappa/\ln(H/z_0)]^2$; although the Nikuradse prefactor c_z is $\sim 10\times$ uncertain, the log law **buffers** it — a $10\times z_0$ uncertainty propagates to only $\sim 1.7\times$ in C_d (validation/synthetic/a3_dimensional_bridge.py, a2_z0_roughness.py; 6+5 tests). (Röthlisberger 1972; Werder et al. 2013; Cuffey
110 & Paterson 2010; Nikuradse 1933; Curl 1966)

4.4 The linear kernel is the small-amplitude limit of the nonlinear system — and the nonlinearity is itself a falsifiable prediction. The §4 projection is exact but *linear*, and a fair objection is that eliminating one variable of a 2×2 system is a linear-algebra identity. We therefore embed it in the **physically nonlinear** Röthlisberger cavity $\leftrightarrow$ channel model — turbulent $\sqrt{p}$ channel discharge and the Glen-law N^3 creep closure (Röthlisberger 1972; Schoof 2010; Hewitt 2013) — and linearise
115 at its steady state. The linearisation recovers *exactly* the overdamped M of §4 (real, negative eigenvalues), so **the linear MZ kernel** $K(\tau) = -ab \cdot e^{-\tau/\tau_2}$ **is the small-amplitude limit**: a plant-and-recover sweep shows the nonlinear surge-lag t^* converging to the linear value as the drainage amplitude $\rightarrow 0$ (relative gap $< 0.5\%$ at a 2 %-of-baseflux flood, rising smoothly with amplitude — Fig. 1). The leading nonlinear correction is not cosmetic; it is driven by the N^3 creep term and yields a falsifiable geophysical prediction. A larger drainage drives water pressure up, $N = p_i - p_w$ down, collapses the N^3 creep
120 closure, keeps the channel open longer, and **lengthens the lag**: the model gives $t^* \approx 0.10yr$ at small floods, rising $\sim 14\%$ to $\approx 0.11yr$ at an 80 %-of-baseflux flood ($dt^*/d(\text{floodfraction}) \approx 0.018yr$). So *bigger floods should have longer surge lags* — a statement the linear kernel cannot make, testable against a drainage-volume-stratified lag population (the USAP-DC 601439 volumes of §5 already span 0.13–2.9 km³). This converts the exact-but-linear certification into the leading term of a controlled, physically-grounded expansion with a testable next-order signature. The prediction is **not a parameter artefact**:
125 across a physical sweep of steady effective pressure, storage and melt-opening efficiency, bigger-floods-lag-longer holds for *every* overdamped configuration and **strengthens toward flotation** (the lag–flood slope rises from $\approx 0.010yr$ at $N^* = 0.55$ to $\approx 0.032yr$ at $N^* = 0.25$), i.e. the effect is largest exactly in the low- N near-flotation regime where surges matter most. (validation/synthetic/hydraulic_nonlinear_kernel.py, 2 tests.)

4.5 The kernel survives spatial resolution: a distributed operator projection whose no-transport limit is the lumped
130 **2×2 .** The other half of the “ 2×2 identity” objection is *spatial*: a lumped two-box model has no geometry, so is the memory kernel an artefact of collapsing the drainage line to a point? We resolve the flowline. Take the linearised cavity $\leftrightarrow$ channel system as two coupled *fields* on a 1-D flowline — $\partial_t s = D_s \partial_{xx} s - s/\tau_1 - aq$ (cavity store s) and $\partial_t q = D_q \partial_{xx} q - U \partial_x q + bs - q/\tau_2$ (the channel q , which *routes water downstream* at speed U and spreads it at D_q ; Röthlisberger 1972; Werder et al. 2013) — and project out the channel *field* exactly (linear Nakajima–Zwanzig). The resolved store obeys a generalized Langevin equation
135 $\dot{s} = A_{ss}s + \int_0^t \mathcal{K}(t-\tau)s(\tau)d\tau + R$ whose kernel $\mathcal{K}(\tau) = A_{sq}e^{A_{qq}\tau}A_{qs}$ is now a *matrix-valued, spatially non-local* operator. On $N = 24$ flowline nodes (validation/synthetic/hydraulic_mz_spatial.py, 6 tests): (i) the projection is **exact** — the GLE transfer operator equals the full resolvent’s resolved block at random complex s , max rel-err 7×10^{-16} ; (ii) the kernel is the eliminated channel operator’s Green’s function (3×10^{-17} vs *expm*) and is spatially non-local with a resolved along-flow range $\ell_{mem} \approx 0.31 \gg \Delta x$; (iii) the channel-eliminated field-GLE reproduces the full PDE trajectory ($1 \times$

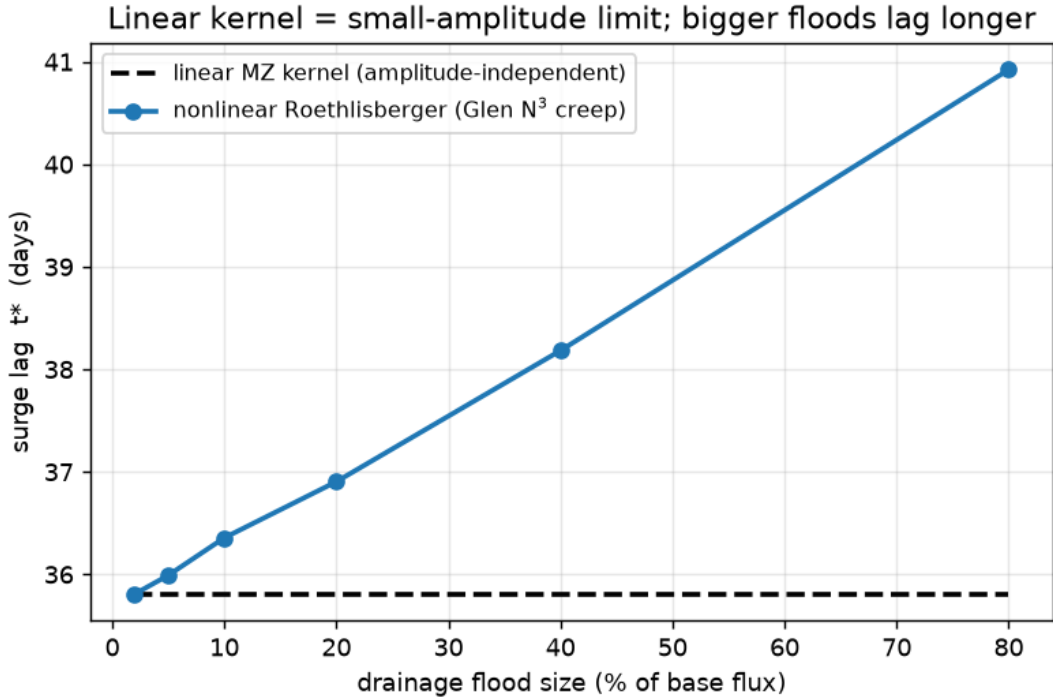


Figure 1. Surge lag t^* versus drainage flood size in the nonlinear R  thlisberger cavity↔channel model. The linear Mori–Zwanzig kernel (dashed) is amplitude-independent and equals the nonlinear lag in the small-flood limit; the Glen-law N^3 creep closure makes the nonlinear lag (circles) rise with flood size — bigger floods lag longer, a falsifiable prediction the linear kernel cannot make.

140 10^{-5}); and (iv) memory necessity is a **dose-response** — the memoryless local closure’s trajectory error grows monotonically with the channel time τ_2 and vanishes as $\tau_2 \rightarrow 0$ (the Markovian limit, now derived *in the distributed system*). Crucially, **the committed lumped 2×2 kernel is exactly this projection’s no-transport, single-node limit**: with $D_q = U = 0$ the operator kernel $\mathcal{K}(\tau)$ is exactly diagonal and its diagonal equals $-abe^{-\tau/\tau_2}$ to 7×10^{-18} . Switching channel transport on moves the kernel mass off-diagonal (off-diagonal fraction $0 \rightarrow 0.99$), so the surge-lag memory acquires a finite **along-flow footprint** with

145 the closed-form decay length $\ell_{mem} = 2D_q/(\sqrt{U^2 + 4D_q/\tau_2} - U)$ (verified against the numeric steady-influence operator to 4×10^{-4} ; *ballistic* limit $\ell_{mem} \rightarrow U\tau_2$ for fast channels, *diffusive* limit $\ell_{mem} \rightarrow \sqrt{D_q\tau_2}$ for $U \rightarrow 0$) — a new, falsifiable, **field-measurable** structural prediction: a drainage’s lagged velocity response is spread along the flowline over ℓ_{mem} , which a memoryless (local) closure sets to zero. This converts the 2×2 “identity” into the spatially-local limit of an exact projection of a genuine distributed (PDE) operator. Scope: still a *linearised* distributed model on a periodic flowline; the fully nonlinear

150 spatially-resolved coupled PDE remains future work.

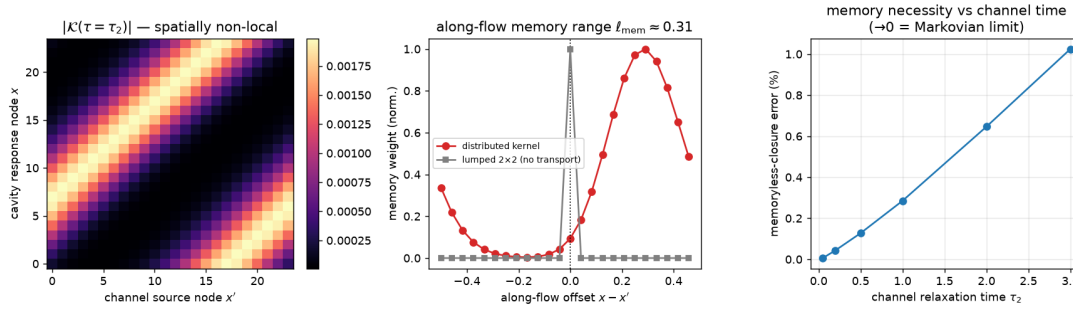


Figure 2. The distributed cavity $\leftrightarrow$ channel memory kernel $\mathcal{K}(\tau = \tau_2)$ (left) is spatially non-local — the off-diagonal band is the downstream channel transport that the lumped 2×2 omits; its along-flow influence profile (centre) has a finite memory range $\ell_{mem} \approx 0.31$ versus the lumped kernel’s zero-range spike; and the memoryless local-closure trajectory error (right) grows with the channel relaxation time τ_2 , vanishing in the Markovian ($\tau_2 \rightarrow 0$) limit.

5 The matched lag test — an honest null

Both halves of a matched test are open: drainage **dates** from CryoSat-2 in-lake elevation (*detect_{drainages}* extracts 7 fill $\rightarrow$ drain events across 5 lakes, including the documented Mercer ≈ 2012.5 drainage), and velocity **response** from ITS_LIVE box-mean surface speed.

lake	base speed (m/yr)	drainages	annual noise floor	quarterly scatter	peak post-drainage
Mac1	422	3	1.0%	2.3%	+0.56σ
Mac2	402	1	1.1%	2.1%	+0.27 σ
Mac3	394	1	1.1%	3.4%	-0.12 σ

Result. No matched lake shows a post-drainage surge above the 2σ floor (peak $+0.56\sigma$, 0/5 significant). The lag value stays *not* promoted to.

Why null is a *stronger* statement than “too noisy”. The modern ITS_LIVE record is well resolved (annual $\approx 1\%$, quarterly $\approx 2\text{--}3\%$), so the limit is not a noise budget. Two distinct, quantifiable limits remain: (1) **temporal aliasing** — the derived lag is sub-annual (t^* baseline ≈ 0.01 yr, p95 ≈ 0.1 yr), finer than the finest *robust* ITS_LIVE bin (quarterly, 0.25 yr); a brief transient is averaged out regardless of scatter (a Nyquist limit); (2) **amplitude bound** — a *sustained* speed-up would survive aliasing as a step, yet the peak anomaly is only $+0.56\sigma$, bounding any sustained surge to $\lesssim 10$ m/yr on ~ 400 m/yr. These map onto the two routes to a verified lag: **sub-annual GPS/GNSS** (EarthScope/POLNET; beats aliasing) **or fast-outlet trunk velocity** (e.g. Byrd, Stearns et al. 2008; steep $v(N)$ sensitivity beats the amplitude floor), paired with that site’s drainage dates.

Vetted forcing obtained. USAP-DC 601439 (Smith et al. 2009 ICESat) yields **58 drainage events across 52 lakes** (median drained volume 0.128 km^3 , max 2.92 km^3 ; implied water flux q_{water} median $\approx 2.5\text{ m}^3/\text{s}$, p95 $\approx 31\text{ m}^3/\text{s}$ — squarely in the literature subglacial-flood range), anchoring the §3/§4 forcing magnitude to real observations.

6 A continental intrusion screen: RTN is the flotation ratio, and its only escape

6.1 Definition. $RTN = (p_{ocean} - p_{atm})/p_w$ on grounded ice: overburden $p_i = \rho_i g H$, ocean head at the bed $p_{ocean} = \rho_w g \cdot d_{base}$ ($d_{base} = \max(0, -bed)$), subglacial water $p_w = \phi p_i$ (so $N_{eff} = (1 - \phi)p_i$), gauge-corrected so the atmosphere cancels ($p_{atm} = 0$). Then $RTN = \rho_w d_{base}/(\phi \rho_i H)$, so ‘RTN>1 $\iff \rho_w d_{base}/(\rho_i H)$

ϕ —the ice is within a factor ϕ of flotation ($N \rightarrow 0$ ’; Schoof 2007; Leguy et al.

2014; Cuffey & Paterson 2010). **RTN is, by construction, a nondimensional form of the flotation/effective-pressure criterion, not an independent predictor.**

6.2 Result on Bedmap2. Over 1.33 M grounded cells (1 km decimated to 3 km, $\phi = 0.9$), the $RTN > 1$ fraction decays monotonically with distance to the grounding line (15.1 % within 5 km $\rightarrow$ 0 % beyond 100 km; median distance 6 km for $RTN > 1$ cells vs 222 km for the rest), and the directional result reproduces on the independent **BedMachine v4** (NSIDC-0756) inversion (median 6 km vs 200 km to the grounding line).

6.3 No skill over thickness-above-flotation (constant ϕ). A direct baseline test on 478,767 grounded Bedmap2 cells confirms $RTN > 1$ is *identical* to the thickness-above-flotation threshold $H_{af}/H < 1 - \phi$ (agreement **100.0000 %**, 0 disagreeing cells, $\phi = 0.8/0.9/0.95$), with $Spearman(RTN, H_{af}/H) = -1.000$ — so at constant ϕ , **RTN adds no skill**; it is that standard quantity, nondimensionalised by ϕ (`validation/external/rtn_baseline_skill.py`).

6.4 The only escape — a data-derived $\phi(x, y)$. Because $RTN = (1 - H_{af}/H)/\phi$ is algebraically forced to track the flotation fraction at constant ϕ , RTN ’s only possible independent signal is ϕ *itself*. A synthetic plant-and-recover (`validation/synthetic/rtn_variable_phi_skill.py`, 5 tests) shows a connectivity-informed $\phi(x, y)$ adds genuine skill over thickness-above-flotation (near-flotation-band $F1$ 0.78 $\rightarrow$ 0.89; AUC 0.995 $\rightarrow$ 0.999), a random ϕ of the same distribution adds none, and the gain rises monotonically with ϕ ’s informativeness. On real Bedmap2, a calibration-free $\phi(x, y)$ from subglacial hydraulic-potential routing (Shreve 1972: priority-flood fill + D8 flow accumulation) reproduces the constant- ϕ anchor exactly and then reorders the intrusion class on $\sim 10^3$ near-grounding-line cells (near-flotation-band $Spearman$ -0.99 vs -1.00), nearly rank-independent of the flotation fraction in-band (`validation/external/rtn_variable_phi_real.py`). The reordering is real but its *correctness* is unvalidated — the routed ϕ is a model field derived from the same geometry — so the decisive test needs a ϕ from an **independent** observation of basal water (radar bed specularity; Schroeder et al. 2013, 2015), scored against a future gridded intrusion survey. We thus present variable- ϕ RTN as a falsifiable screen with one clearly-stated open data step, not a verified forecast.

6.5 Intrusion residence number Ro . The level-set advance speed of the $RTN = 1$ front is $v_{kin} = A \cdot dH/dt$; the residence number $Ro = v_{kin} \cdot \tau_{hyd}/\ell$ (regime boundary $Ro = 1$ at the critical residence $\tau_{crit} = \ell/v_{kin}$) predicts whether intrusion is thinning-paced ($Ro \approx 1$) or hydraulic-limited ($Ro \gg 1$). For the runaway-tail cells ($A = 0.70 \text{ km/m}$, $dH/dt = 1.5 \text{ m/yr}$) ~ 98 % of the §4 residence band lies below $\tau_{crit} \approx 1.9 \text{ yr} \Rightarrow$ predicted **thinning-paced**; a DInSAR v_{obs} would place it on the curve (`validation/synthetic/intrusion_residence_number.py`, 7/7 tests).

6.6 One state variable behind three thresholds (a derived unifier). Height above flotation $h_{af} = H - H_f$ (with $H_f = (\rho_w/\rho_i)d_{base}$; Schoof 2007) ties the three thresholds this framework treats separately into one. The ocean-connected effective

pressure is $N = \rho_i g h_{af}$ *exactly* (so $N = 0 \iff h_{af} = 0 \iff$ flotation); the $RTN = 1$ intrusion surface is $h_{af} = H(1 - \phi)$, which $\rightarrow 0$ as $\phi \rightarrow 1$ (the intrusion surface *is* the flotation surface once subglacial water reaches overburden); and the sliding-law $|s_N|$ pole (Paper 4b) sits at $h_{af}^c = N_c / (\rho_i g) > 0$, a few metres *above* geometric flotation. A thinning stream therefore
205 reaches the drag-side fold (the velocity early-warning) *first* and ungrounds ($h_{af} \rightarrow 0$) only later — a buffer set by N_c . RTN, the Schoof saddle-node, and the s_N pole are thus three readings of one state variable, not independent diagnostics (verified on a synthetic thinning sweep: $N = \rho_i g h_{af}$ to machine zero, $h_{af}^c = 6.67$ m; `nr10_flotation_unifier`).

7 Estimator calibration (synthetic plant-and-recover)

Every real-data estimator has a synthetic null/recovery harness exercised in `pytest`:

- 210 – **Lag estimator is kernel-shape-generic** — plant one target lag with five markedly different causal kernels (Gamma $k = 2/4$, log-normal, bi-exponential, raised-cosine); `estimate_lag` recovers it to ≤ 10 % for every shape; a delta-kernel control returns 0.
- **RTN classifier / ϕ -area inversion / variable- ϕ skill** — planted thresholds and connectivity recovered without bias (`rtn_synthetic`, `rtn_phi_synthetic`, `rtn_variable_phi_skill`).
- 215 – **Grounding-line migration level-set law is exact** — $|dH/dt|/|\nabla m|$ reproduced to rel-err 5×10^{-16} (1-D) / 4×10^{-14} (tilted 2-D).
- **MZ projection** — the five §4 properties to machine precision (`hydraulic_mz_projection_synthetic`, 6/6).

8 Discussion and limitations

What this is. A machine-precision certification of the cavity $\leftrightarrow$ channel kernel’s exact MZ structure (the central positive result);
220 a motivated null on a literal thermal sliding kernel; an honest matched-lag null with a precise diagnosis and a concrete unblock path; and a flotation-ratio intrusion *screen* whose only genuine degree of freedom (a data-derived $\phi(x, y)$) is isolated and partially tested.

What this is not. No verified surge-lag *value* (the matched test is null; the value is to order of magnitude only). The §4.4 nonlinear extension shows the linear kernel is the small-amplitude limit of the nonlinear *lumped* Röthlisberger system and
225 predicts an amplitude-dependent lag, and the §4.5 distributed extension certifies the kernel as an exact spatially non-local operator projection of the linearised flowline PDE; only the fully *nonlinear spatially-resolved* coupled GlaDS/Röthlisberger PDE remains future work. No RTN precision/recall (no gridded survey). The physical distinctness of ϕ from the Leray pressure remains unproven.

Open gates, each with a concrete unblock. (1) Matched lag $\rightarrow$ sub-annual GPS/GNSS or fast-outlet trunk velocity, paired
230 with drainage dates. (2) The *linear* distributed projection (§4.5) and the *lumped* nonlinear extension (§4.4) are done; the

remaining gate is the fully *nonlinear spatially-resolved* coupled GlaDS/Röthlisberger PDE. (3) RTN skill $\rightarrow$ a $\phi(x, y)$ from an independent basal-water observable (specularity) + a gridded intrusion survey.

9 Reproduce

```

pip install -r requirements.txt
235 python glaciers/validation/external/run_sliding_real.py --with-velocity Mac1 # $3 tau_ice +
python -m glaciers.validation.external.lag_fit_real # $5 matched-la
python glaciers/validation/external/run_usapdc_lakes.py # $5 vetted for
pytest glaciers/tests/test_validation_synthetic.py -v # $4 MZ project
python glaciers/validation/external/rtn_baseline_skill.py --stride 5 # $6.3 RTN==flo
240 python glaciers/validation/synthetic/rtn_variable_phi_skill.py # $6.4 variable
python glaciers/validation/external/rtn_variable_phi_real.py --stride 10 # $6.4 hydrauli
python glaciers/validation/synthetic/intrusion_residence_number.py # $6.5 residenc
python glaciers/validation/synthetic/hydraulic_nonlinear_kernel.py # $4.4 nonline
python glaciers/validation/synthetic/hydraulic_mz_spatial.py # $4.5 distrib

```

245 *Code and data availability.* All analysis code and derived products are in the public repository at <https://github.com/abdh0saifelden2-debug/K>, and the results regenerate from the scripts above. The study uses the following open datasets:

- Bedmap2 ice thickness (Fretwell et al., 2013, <https://doi.org/10.5194/tc-7-375-2013>).
- Independent ice thickness and active-lake inventory: USAP-DC 601470 (Stubblefield et al., 2021, <https://doi.org/10.15784/601470>; ice thickness after Arthern et al., 2015; active lakes after Siegfried & Fricker, 2018).
- 250 – Active subglacial-lake drainage volumes: USAP-DC 601439 (Smith et al., 2012, <https://doi.org/10.15784/601439>; ICESat inventory of Smith et al., 2009).
- BedMachine Antarctica v4 (Morlighem, 2025, <https://doi.org/10.5067/POJQI54A45HX>; see Morlighem et al., 2020) for the independent continental RTN screen.
- ITS_LIVE surface velocities (Gardner et al., 2025, <https://doi.org/10.5067/JQ6337239C96>; see Gardner et al., 2018) and CryoSat-2
- 255 lake-drainage dates.

USAP-DC data were obtained from the U.S. Antarctic Program Data Center (<https://www.usap-dc.org>); BedMachine and ITS_LIVE from the NASA NSIDC DAAC.

Author contributions. A. Saifelden conceived the study, performed all computations and analysis, and wrote the manuscript.

Competing interests. The author declares that there are no competing interests.

- Arthern, R. J., Hindmarsh, R. C. A., & Williams, C. R. (2015). Flow speed within the Antarctic ice sheet and its controls inferred from satellite observations. *Journal of Geophysical Research: Earth Surface* 120(7), 1171–1188. doi:10.1002/2014JF003239
- Cuffey, K. M., & Paterson, W. S. B. (2010). *The Physics of Glaciers*, 4th ed. Academic Press, Amsterdam.
- Curl, R. L. (1966). Scallops and flutes. *Transactions of the Cave Research Group of Great Britain* 7, 121–160.
- 265 Fretwell, P., Pritchard, H. D., Vaughan, D. G., et al. (2013). Bedmap2: improved ice bed, surface and thickness datasets for Antarctica. *The Cryosphere* 7, 375–393. doi:10.5194/tc-7-375-2013
- Gardner, A. S., Moholdt, G., Scambos, T., et al. (2018). Increased West Antarctic and unchanged East Antarctic ice discharge over the last 7 years. *The Cryosphere* 12, 521–547. doi:10.5194/tc-12-521-2018
- Gardner, A., Fahnestock, M., Greene, C. A., Kennedy, J. H., Liukis, M., Lopez, L., & Scambos, T. (2025). MEaSURES ITS_LIVE Regional
270 Glacier and Ice Sheet Surface Velocities, Version 2 (NSIDC-0776) [Data set]. *NASA NSIDC DAAC*. doi:10.5067/JQ6337239C96
- Hewitt, I. J. (2013). Seasonal changes in ice sheet motion due to melt water lubrication. *Earth and Planetary Science Letters* 371–372, 16–25. doi:10.1016/j.epsl.2013.04.022
- Leguy, G. R., Asay-Davis, X. S., & Lipscomb, W. H. (2014). Parameterization of basal friction near grounding lines in a one-dimensional ice sheet model. *The Cryosphere* 8, 1239–1259. doi:10.5194/tc-8-1239-2014
- 275 Morlighem, M., Rignot, E., Binder, T., et al. (2020). Deep glacial troughs and stabilizing ridges unveiled beneath the margins of the Antarctic ice sheet. *Nature Geoscience* 13, 132–137. doi:10.1038/s41561-019-0510-8
- Morlighem, M. (2025). MEaSURES BedMachine Antarctica, Version 4 (NSIDC-0756) [Data set]. *NASA National Snow and Ice Data Center DAAC*. doi:10.5067/POJQI54A45HX
- Mori, H. (1965). Transport, collective motion, and Brownian motion. *Progress of Theoretical Physics* 33, 423–455. doi:10.1143/PTP.33.423
- 280 Nakajima, S. (1958). On quantum theory of transport phenomena. *Progress of Theoretical Physics* 20, 948–959. doi:10.1143/PTP.20.948
- Nikuradse, J. (1933). Strömungsgesetze in rauhen Röhren. *VDI-Forschungsheft* 361. (English transl. NACA TM 1292, 1950.)
- Röthlisberger, H. (1972). Water pressure in intra- and subglacial channels. *Journal of Glaciology* 11, 177–203. doi:10.3189/S0022143000022188
- Schoof, C. (2007). Ice sheet grounding line dynamics: Steady states, stability, and hysteresis. *Journal of Geophysical Research: Earth Surface*
285 112, F03S28. doi:10.1029/2006JF000664
- Schoof, C. (2010). Ice-sheet acceleration driven by melt supply variability. *Nature* 468, 803–806. doi:10.1038/nature09618
- Schroeder, D. M., Blankenship, D. D., & Young, D. A. (2013). Evidence for a water system transition beneath Thwaites Glacier, West Antarctica. *Proceedings of the National Academy of Sciences* 110(30), 12225–12228. doi:10.1073/pnas.1302828110
- Schroeder, D. M., Blankenship, D. D., Raney, R. K., & Grima, C. (2015). Estimating subglacial water geometry using radar bed
290 echo specularly: Application to Thwaites Glacier, West Antarctica. *IEEE Geoscience and Remote Sensing Letters* 12(3), 443–447. doi:10.1109/LGRS.2014.2337878
- Shreve, R. L. (1972). Movement of water in glaciers. *Journal of Glaciology* 11, 205–214. doi:10.3189/S002214300002219X
- Siegfried, M. R., Fricker, H. A., Carter, S. P., & Tulaczyk, S. (2016). Episodic ice velocity fluctuations triggered by a subglacial flood in West Antarctica. *Geophysical Research Letters* 43, 2640–2648. doi:10.1002/2016GL067758
- 295 Siegfried, M. R., & Fricker, H. A. (2018). Thirteen years of subglacial lake activity in Antarctica from multi-mission satellite altimetry. *Annals of Glaciology* 59, 42–55. doi:10.1017/aog.2017.36

- Siegfried, M. R., & Fricker, H. A. (2021). Illuminating active subglacial lake processes with ICESat-2 laser altimetry. *Geophysical Research Letters* 48, e2020GL091089. doi:10.1029/2020GL091089
- Smith, B. E., Fricker, H. A., Joughin, I. R., & Tulaczyk, S. (2009). An inventory of active subglacial lakes in Antarctica detected by ICESat (2003–2008). *Journal of Glaciology* 55, 573–595. doi:10.3189/002214309789470879
- Smith, B., Fricker, H., Joughin, I., & Tulaczyk, S. (2012). Antarctic Active Subglacial Lake Inventory from ICESat Altimetry. *U.S. Antarctic Program Data Center*. doi:10.15784/601439
- Stearns, L. A., Smith, B. E., & Hamilton, G. S. (2008). Increased flow speed on a large East Antarctic outlet glacier caused by subglacial floods. *Nature Geoscience* 1, 827–831. doi:10.1038/ngeo356
- Stubblefield, A. G., Arthern, R. J., Kingslake, J., & Siegfried, M. R. (2021). Antarctic ice thickness, slipperiness, and subglacial lake locations. *U.S. Antarctic Program Data Center*. doi:10.15784/601470
- Werder, M. A., Hewitt, I. J., Schoof, C. G., & Flowers, G. E. (2013). Modeling channelized and distributed subglacial drainage in two dimensions. *Journal of Geophysical Research: Earth Surface* 118, 2140–2158. doi:10.1002/jgrf.20146
- Wingham, D. J., Francis, C. R., Baker, S., et al. (2006). CryoSat: A mission to determine the fluctuations in Earth’s land and marine ice fields. *Advances in Space Research* 37, 841–871. doi:10.1016/j.asr.2005.07.027
- Zwanzig, R. (1960). Ensemble method in the theory of irreversibility. *The Journal of Chemical Physics* 33, 1338–1341. doi:10.1063/1.1731409
- Zwanzig, R. (1973). Nonlinear generalized Langevin equations. *Journal of Statistical Physics* 9, 215–220. doi:10.1007/BF01008729